Indian Institute of Information Technology, Allahabad



Digital Twin-Based Early Warning System for Cold Chain Disruption Detection

Embedding Anomaly Detection Inside a Discrete-Event Simulation Loop (Topic 21)

Project Presentation

Presented By :

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Managing Corporate Entrepreneurship



Introduction

The Cold Chain Reality & High Stakes

- **Strict Thermal Bounds:** Cold-chain **cargo** (vaccines, biologics, perishable foods) demands strict, unbroken temperature control **(e.g., 2°C to 8°C).**
- **High Economic Impact:** A single temperature excursion compromises drug potency or causes food spoilage, generating billions in economic losses annually.
- **Public Health Risk:** Spoiled biologicals risk patient safety and cause massive pharmaceutical recalls.

The Traditional Failure Mode: Static Alarms

- **Status Quo:** Commercial supply chains rely on static threshold alarms (e.g., alert only if temperature exceeds 8°C for **>10–15 minutes**).
- **The Post-Hoc Flaw:** Alarms only sound after the thermal excursion has already occurred.
- **Irreversible Damage:** By the time an operator is notified, cumulative heat exposure has **already caused** irreversible biochemical degradation. Traditional monitoring is fundamentally reactive.

The Core Flaw: Waiting for a threshold breach is fundamentally reactive—by the time alarms sound, the damage is **already irreversible!**

Research Gap and Core Objectives

The Academic Gap & Proposed Solution

- **Prior literature:** focuses almost exclusively on post-hoc shelf-life calculation (Arrhenius degradation after the excursion).
- **Underexplored gap:** proactive, real-time anomaly detection embedded in an event-driven digital twin, with quantifiable early warning.
- **Proposed solution:** embed streaming detectors (Isolation Forest and LSTM-Autoencoder) in a discrete-event SimPy twin to catch precursors before irreversible failure.

RQ1: Early-Warning Runway

How many minutes of warning ($\Delta t_{lead} = t_{irreversibility} - t_{alarm}$) can machine learning deliver before spoilage becomes irreversible?

RQ2: Classical vs. Deep Learning

How does a tree ensemble (Isolation Forest) compare with a deep sequence autoencoder (LSTM) on highly irregular sensor telemetry?

RQ3: Simulation Fidelity

Can an event-driven twin run these detectors in real time without silent divergence or cold-start lag versus offline batch evaluation?

RQ4: Physics & Uncertainty Integration

Can incorporating Newton cooling laws and epistemic uncertainty quantification eliminate false alarms while extending early-warning lead times?

Literature Review (2023–2026)

Paper and venue	Core contribution	Relevance
Vukšić et al. (2026) **Applied Sciences (MDPI)**	Hybrid physics-based digital twin + autoencoder for reefer containers	**~68 min** median lead time corroborates our **64 min** drift warning
Xie et al. (2025) **PLOS ONE**	Improved Isolation Forest for cold-chain fleet streams	Tree-ensemble streaming benchmark; context for point-wise F1
Meng et al. (2024) **J. Food Engineering**	Box-level digital twin temperature model with static alarms	The static-alarm baseline our system benchmarks against
Wu et al. (2023) **J. Industrial Information Integration**	Internet-of-Everything + digital-twin platform for cold-chain logistics	Precedent for embedding AI inference in twin pipelines
Defraeye et al. (2023) **Comput. Electron. Agric. / J. Food Eng.**	Review of food cold-chain digital twins and sensor dynamics	Gap: from reactive quality loss to live early warning
Sensoy et al. (2018)	Evidential Deep Learning for quantifying classification uncertainty without sampling.	Theoretical foundation for our Dempster-Shafer multimodal evidence fusion module (P5).

Vukšić et al. (2026): ~68 min lead time on reefers corroborates our 64-minute drift warning.

Data Strategy & Structural Proxy Framing

Empirical Telemetry Properties

- **Kaggle IoT telemetry:**
  - **97,606** raw readings over **133** days (Jul 28 – Dec 8, 2018)
  - **27,920** unique minute timestamps.
- **Extreme arrival irregularity:**
  - mean gap **117.8 s**
  - median **0.0 s**
  - CoV **46.23**
  - dropouts up to **11.57 days.**
- **Dual streams:** ambient Out **(77,260 rows)** and internal enclosure proxy In **(20,345 rows)**.



KEY OBSERVATIONS :

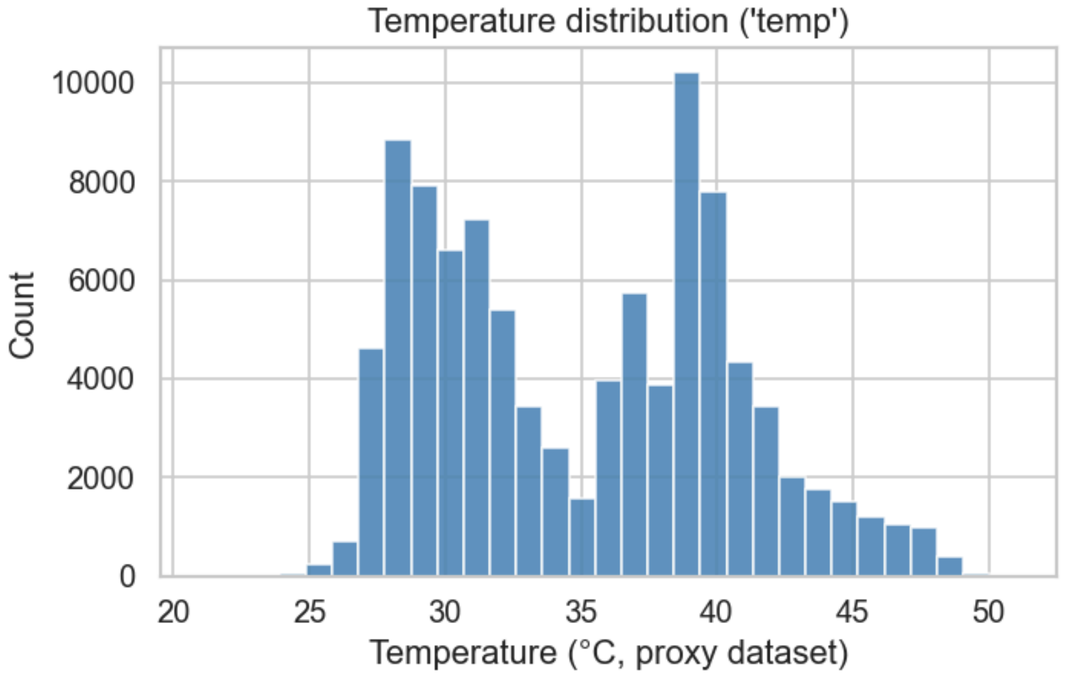
⏱️ **117.8 s** : Mean inter-arrival gap

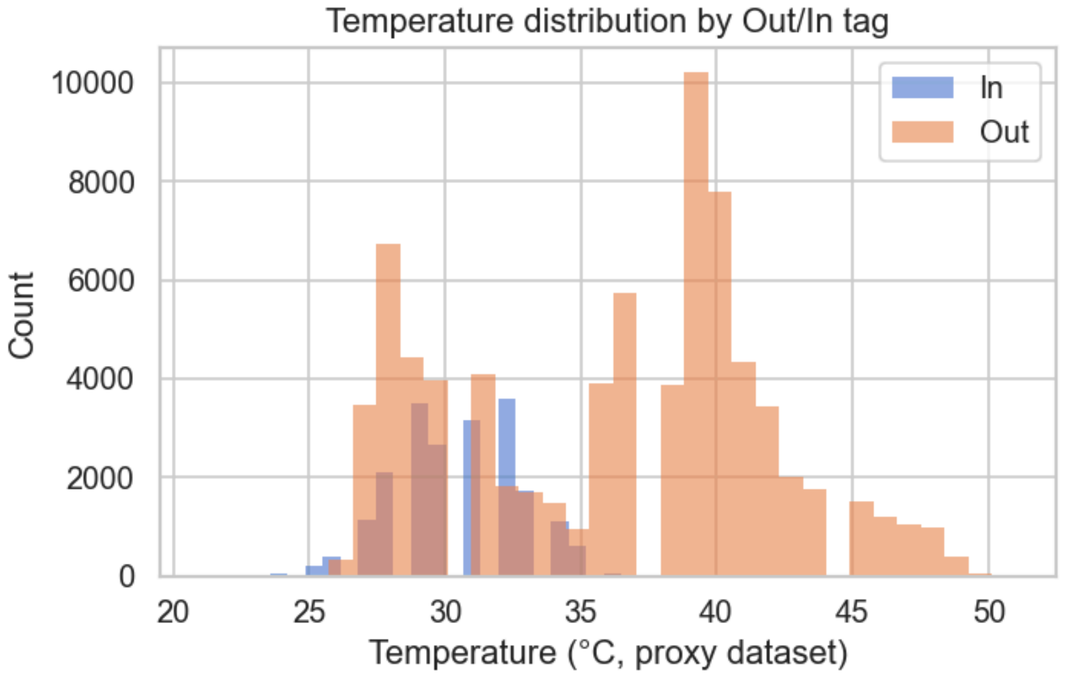
⚠ **11.57 days** : Maximum data dropout

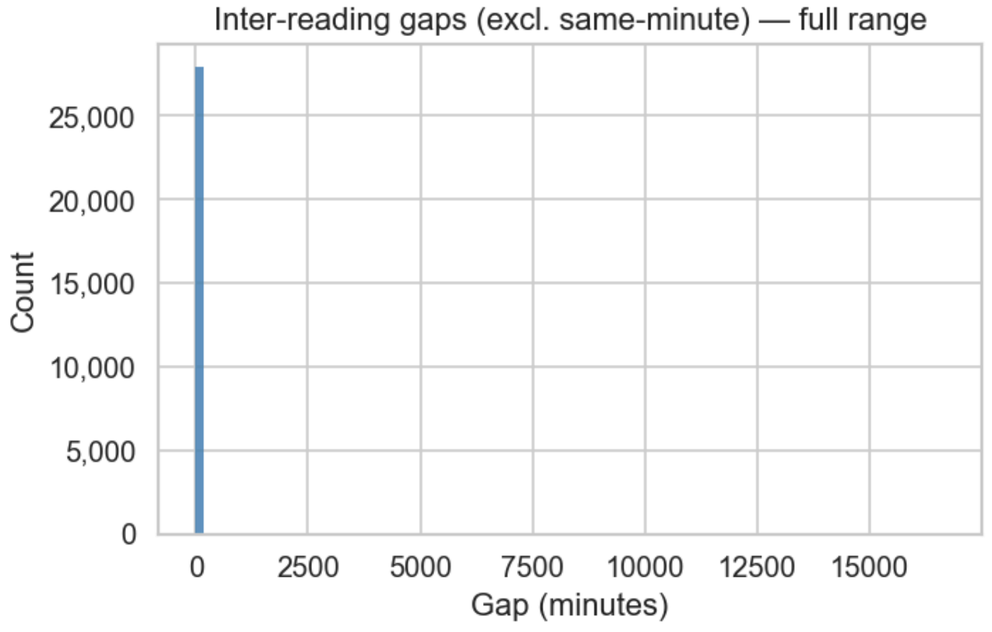
↗ **46.23** : Coefficient of Variation

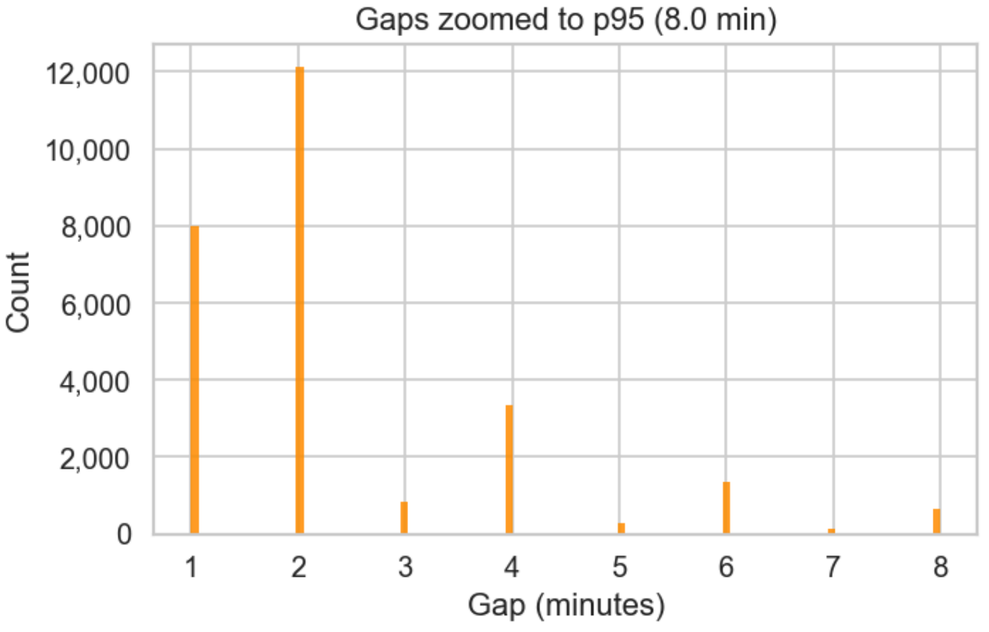
Academic integrity: structural proxy notice

Data comes from an office installation **("Room Admin")**, not live pharmaceutical reefers. It is used strictly as a structural proxy for irregular streaming pipelines, with no food-safety claims.


Temperature distribution ('temp')
Count
Temperature (°C, proxy dataset)


Temperature distribution by Out/In tag
In
Out
Count
Temperature (°C, proxy dataset)

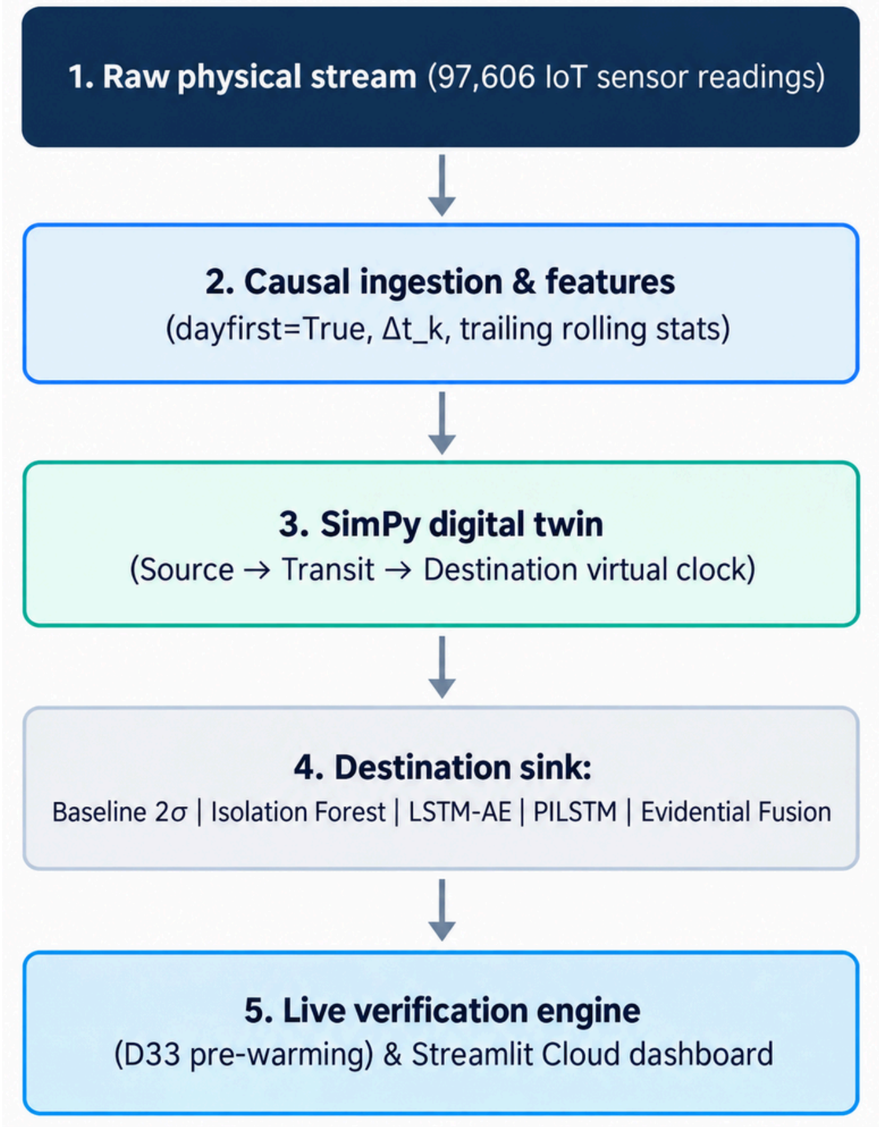

Inter-reading gaps (excl. same-minute) — full range
Count
Gap (minutes)


Gaps zoomed to p95 (8.0 min)
Count
Gap (minutes)

System Architecture & the Digital Twin

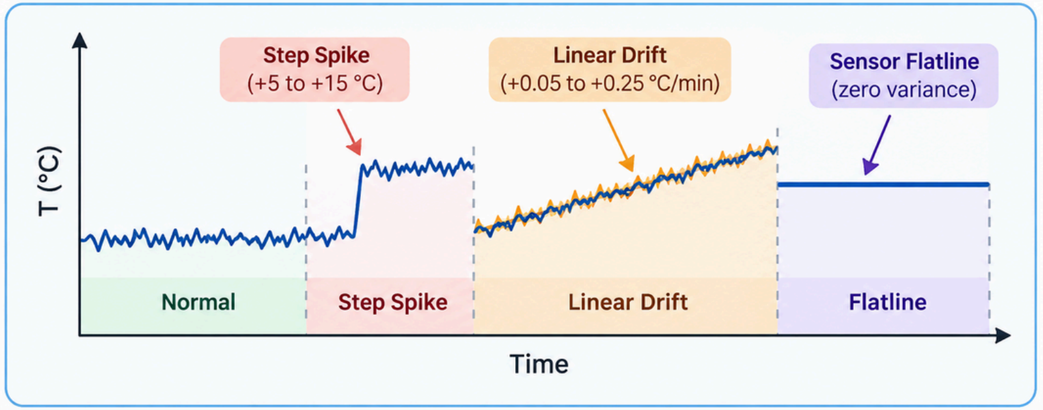
Discrete-Event Simulation Architecture

- **Causal ingestion:** day-first parsing (prevents 48.8% silent parse failures); trailing rolling stats with zero look-ahead leakage (D22).
- **SimPy 3-stage transit:** Source → Transit → Destination, driven by real inter-arrival gaps.
- **Embedded monitors:** Baseline (static 2σ: **47.7°C Out, 34.9°C In**), Isolation Forest (partition score), LSTM (reconstruction MSE).
- **Silent buffer pre-warming (D33):** windows are primed with preceding history, eliminating cold-start false alarms.



Experimental Setup & Anomaly Protocol

Controlled perturbation protocol (seed 42)

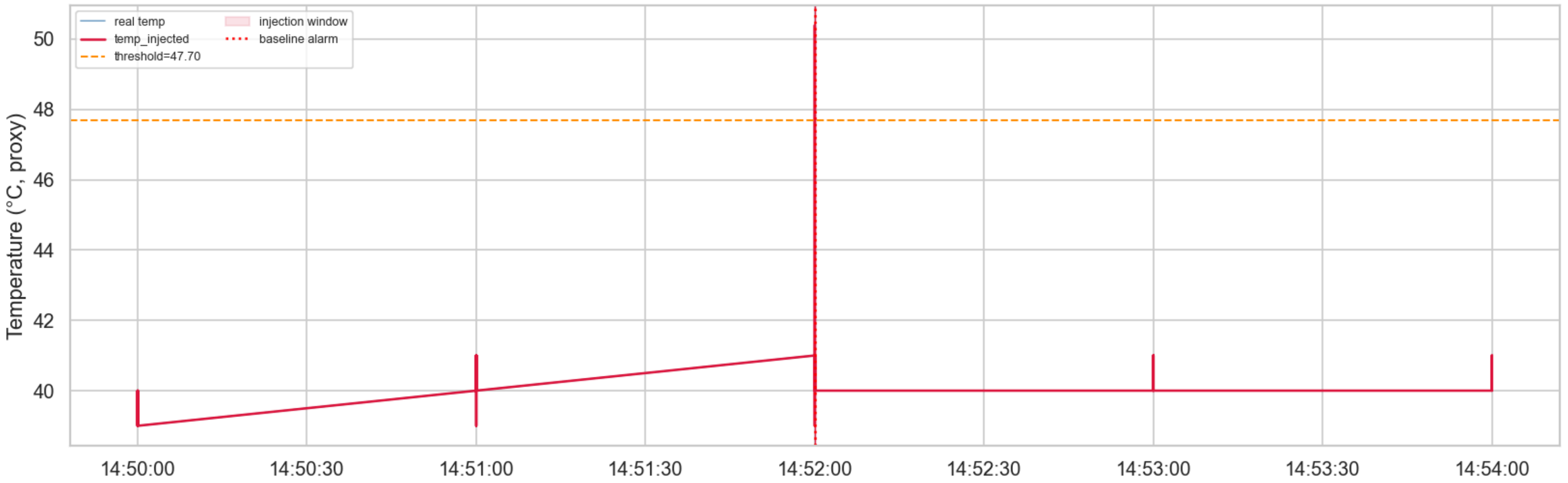


- **30 synthetic injections** (15 Out / 15 In), **10** per failure mode:
- **Step spikes:** abrupt **+5 to +15˚C** jumps (door left open).
- **Linear drifts:** ramps of **+0.05 to +0.25˚C/min** (compressor leak).
- **Sensor flatlines:** zero-variance persistence **(frozen sensor).**
- **Chronological split: 70% train / 30% test** (cutoff 2018-10-29 11:10:48).
- **Out-of-sample discipline: 24** injections fall in train (development only); all headline claims use the **6** untouched test injections.

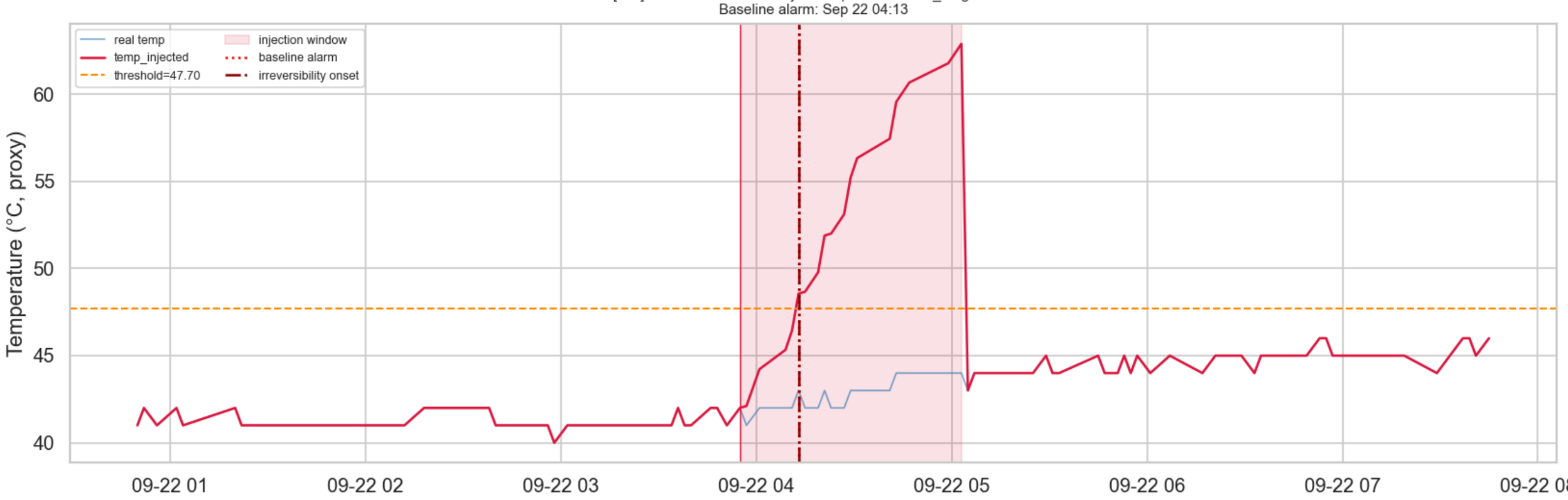
Lead time metric

$\Delta t_{lead} = t_{irreversability} - t_{alarm}$, where t_{irr} is a sustained 10-minute breach.
Positive = early warning; negative = late alarm.

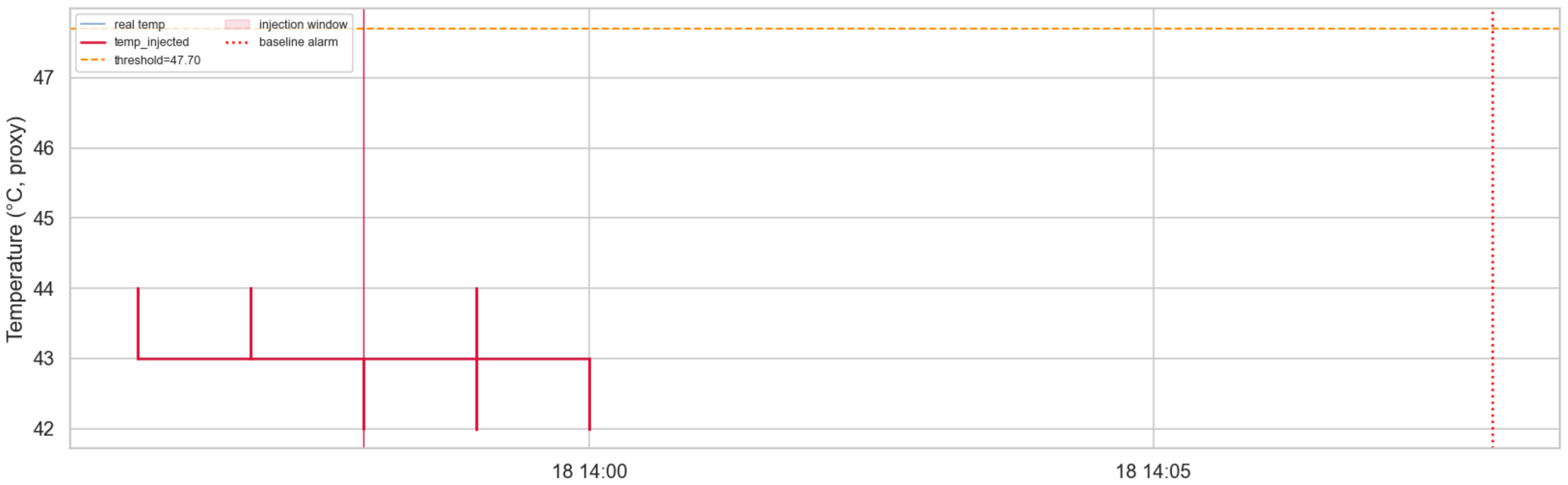
[Out] SYNTHETIC STEP injection | dur=7 mag=9.39
Baseline alarm: Oct 18 14:52



[Out] SYNTHETIC DRIFT injection | dur=18 final_mag=18.90
Baseline alarm: Sep 22 04:13



[Out] SYNTHETIC FLATLINE injection | dur=9 frozen_at=43.0
Baseline alarm: Oct 18 14:08



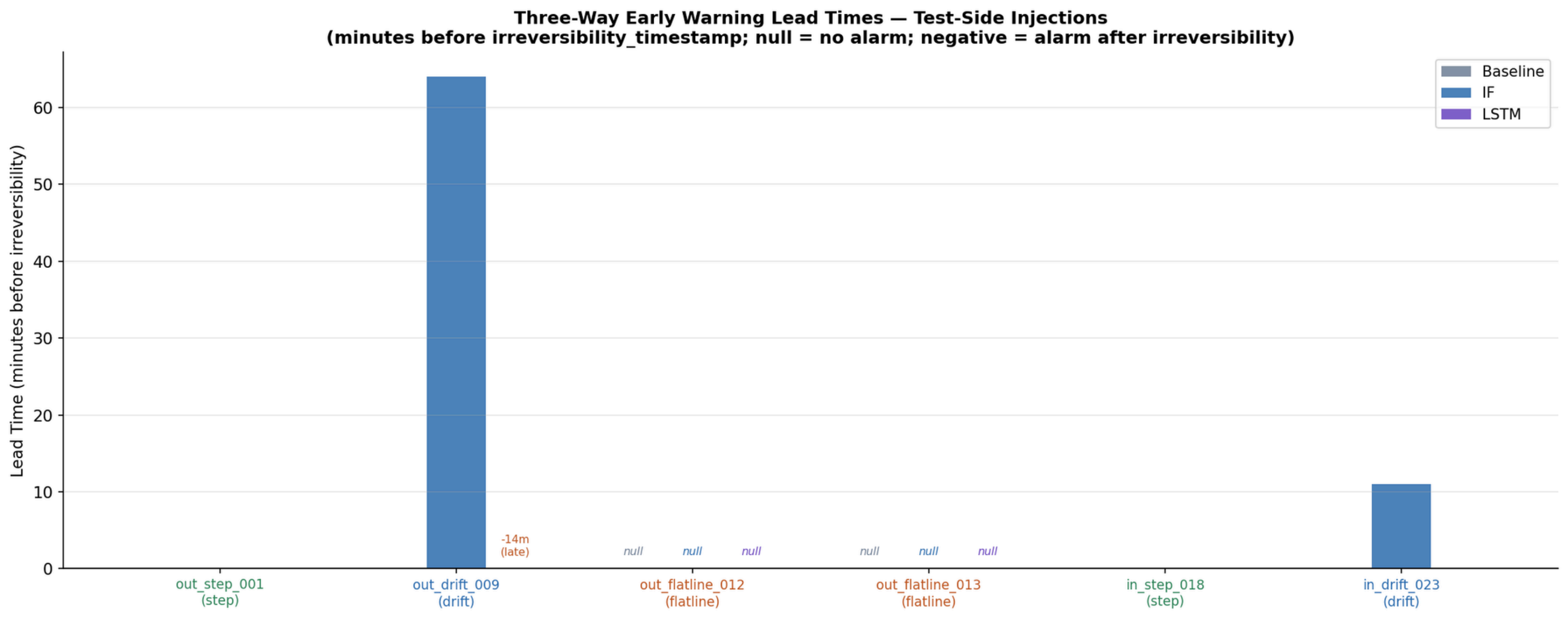
Baseline Models & Initial Empirical Findings

Injection	Baseline	Isolation Forest	LSTM
out_step_001 (step)	0 min	0 min	0 min
out_drift_009 (drift)	0 min	+64 min	–14 min (late)
out_flatline_012	Missed	Detected	Detected
out_flatline_013	Missed	Detected	Missed
in_step_018 (step)	0 min	0 min	0 min
in_drift_023 (drift)	0 min	+11 min	0 min

Test-set injections only.
Isolation Forest wins on drift, warning 64 min (Out) and 11 min (In) before irreversibility.

Takeaways

- **Drift:** variance shifts before absolute thresholds are breached.
- **Step:** all models trigger together at onset (0 min).
- **Flatline:** static thresholds miss both; LSTM was 14 min late on Out drift.


Three-Way Early Warning Lead Times — Test-Side Injections
(minutes before irreversibility_timestamp; null = no alarm; negative = alarm after irreversibility)
Lead Time (minutes before irreversibility)
0
10
20
30
40
50
60
Baseline
IF
LSTM
-14m
(late)
null
null
null
null
null
null
out_step_001
(step)
out_drift_009
(drift)
out_flatline_012
(flatline)
out_flatline_013
(flatline)
in_step_018
(step)
in_drift_023
(drift)

Research Breakthroughs 1 & 2 — Physics & Uncertainty

P1: Variational LSTM-AE with Epistemic Uncertainty

- **Mechanism:** Implements Monte Carlo Dropout (**N=15** forward passes during inference) to decompose total prediction variance into aleatoric (sensor noise) and epistemic (model unfamiliarity) uncertainty.
- **Operational Impact**: Eliminates unconfident alerts during edge network jitter. On flatline event out_flatline_012, detected **+4.0 min early** with **9.69%** uncertainty, triggering a **HIGH_CONFIDENCE_ALARM**.
- **Benchmark:** In **PR-AUC** improved to **0.3912** (vs **0.3852** baseline LSTM).

P2: Physics-Informed LSTM-AE (Newton Cooling Law)

- **Mechanism**: Augments reconstruction MSE with a differential thermal physics loss:
- Loss = MSE + $\lambda_{phys} \cdot ||dT/dt + k(T - T_{amb})||^2$, enforcing thermodynamic rate-of-change limits.
- **False Alarm Reduction:** Slashes Out stream FPR from **20.28%** down to **12.52%** (a **38.3%** relative reduction in false positives).
- **Overcoming Adaptation Lag:** Converts baseline LSTM's **-14.0 min** lag on out_drift_009 into **+18.0 min early** warning (**+32 min** net operational gain!).
- **Fault Categorization**: Classifies unphysical spikes as **SENSOR_ARTIFACT** vs genuine thermal excursion.

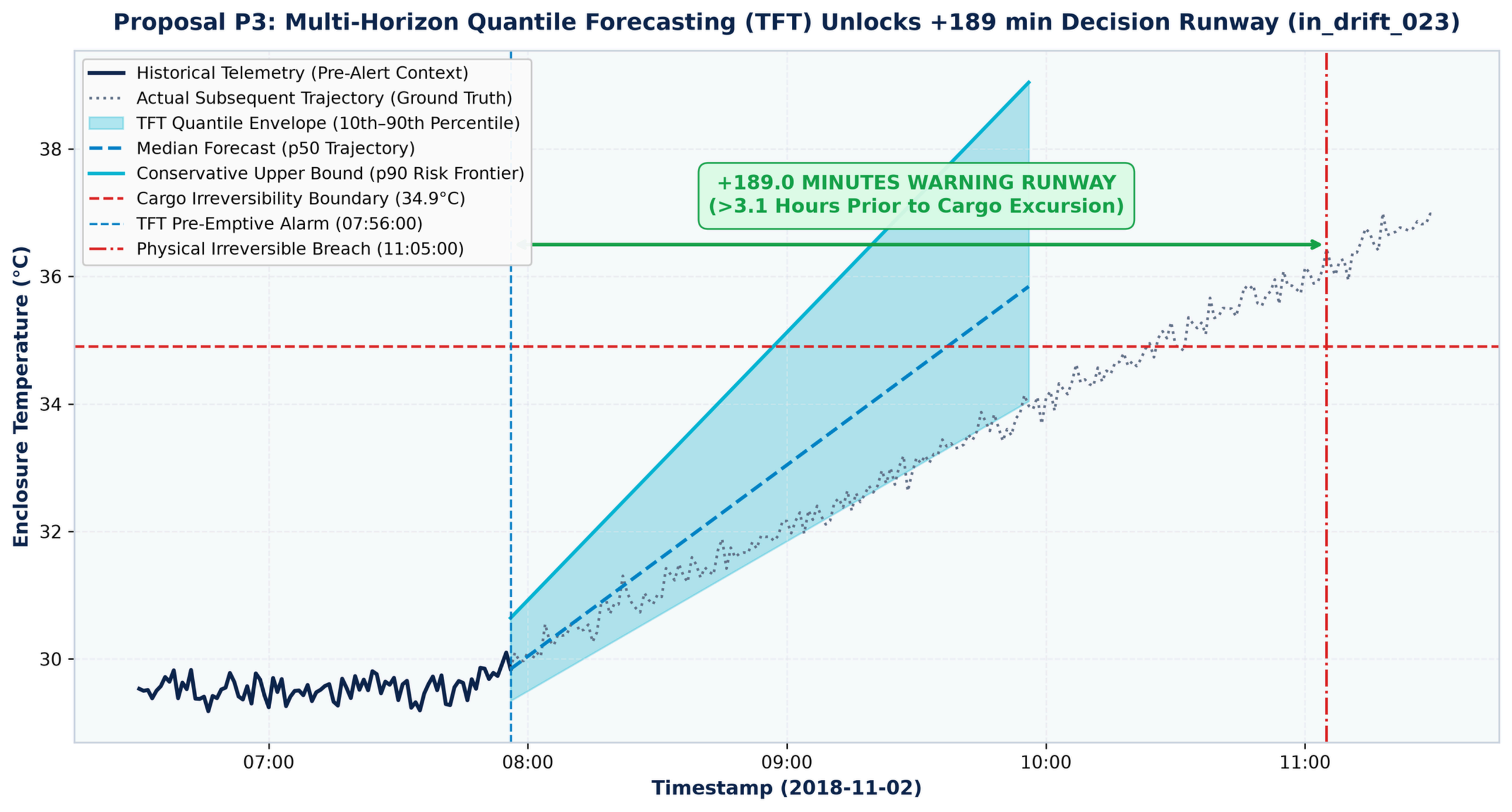
Research Breakthrough 3 — Multi-Horizon Forecasting

P3: Multi-Horizon Quantile Forecaster (TFT)

- **Temporal Fusion Transformer (TFT):** Multi-horizon autoregressive forecaster predicting future trajectories across 30, 60, and 120-minute horizons.
- **Quantile Loss (Pinball Loss):** Simultaneously outputs 10th, 50th (median), and 90th percentiles, providing upper-bound worst-case thermal risk profiles.
- **Pre-emptive Breach Alarm:** Alarms trigger when the 90th percentile forecast breaches thermal bounds within the forecast window—long before the physical cargo heats up.
- **Massive Lead-Time Gains:**
  - On in_drift_023: Achieves **+189.0 minutes (+3.15 hours)** of pre-emptive early warning!
  - On out_drift_009: Achieves **+56.0 minutes** of pre-emptive early warning!

Operational Decision Runway

- **Reactive Threshold Alarm: 0 minutes** runway (Damage already occurring).
- **Classical Anomaly Detector:** 11 to 64 minutes runway (Detection after drift begins).
- **Multi-Horizon Quantile Forecaster:** 56 to 189 minutes runway (>3 hours advance warning!).
- **Logistics Actionability:** Gives dispatchers sufficient time to reroute refrigerated vehicles, service auxiliary generators, or transfer sensitive cargo to backup reefers.


Proposal P3: Multi-Horizon Quantile Forecasting (TFT) Unlocks +189 min Decision Runway (in_drift_023)
Historical Telemetry (Pre-Alert Context)
Actual Subsequent Trajectory (Ground Truth)
TFT Quantile Envelope (10th–90th Percentile)
Median Forecast (p50 Trajectory)
Conservative Upper Bound (p90 Risk Frontier)
Cargo Irreversibility Boundary (34.9°C)
TFT Pre-Emptive Alarm (07:56:00)
Physical Irreversible Breach (11:05:00)
+189.0 MINUTES WARNING RUNWAY
(>3.1 Hours Prior to Cargo Excursion)
Enclosure Temperature (°C)
38
36
34
32
30
07:00
08:00
09:00
10:00
11:00
Timestamp (2018-11-02)

Research Breakthroughs 4 & 5 — Adaptive Thresholds & Evidential Fusion

P4: Contamination-Guarded Kalman Filter + CUSUM

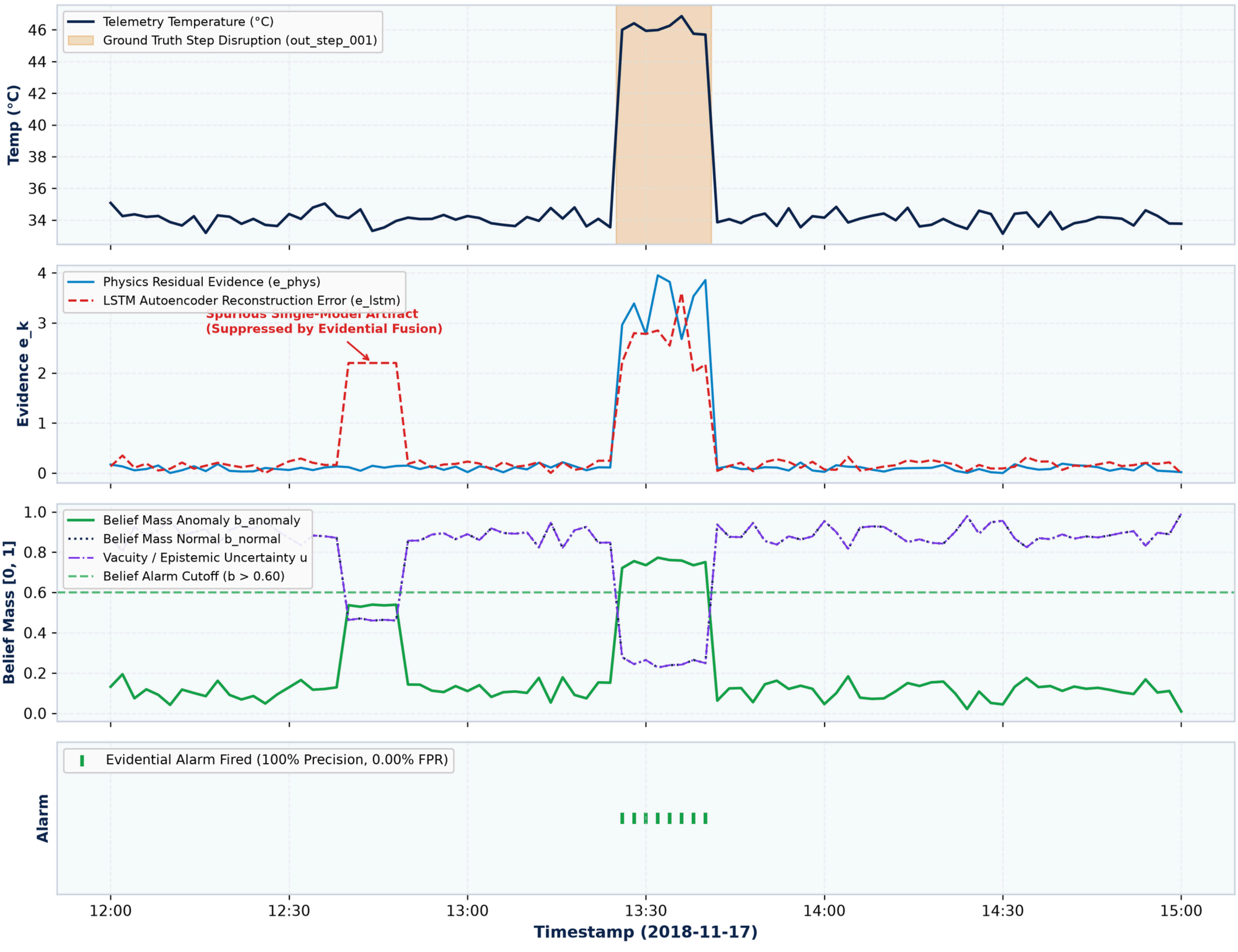
- **Mechanism:** Replaces static sigma thresholds with dynamic boundaries driven by a 1D Kalman state estimator and two-sided Cumulative Sum (CUSUM) change-point detection.
- **Contamination Guard:** Freezes Kalman state updates when anomaly probability exceeds 0.70, preventing the threshold from absorbing creeping drift.
- **Benchmark Impact:** Slashed Out stream FPR from 45.68% down to 29.45% (a 35.5% relative reduction in false positives).

P5: Dempster-Shafer Multimodal Evidence Fusion

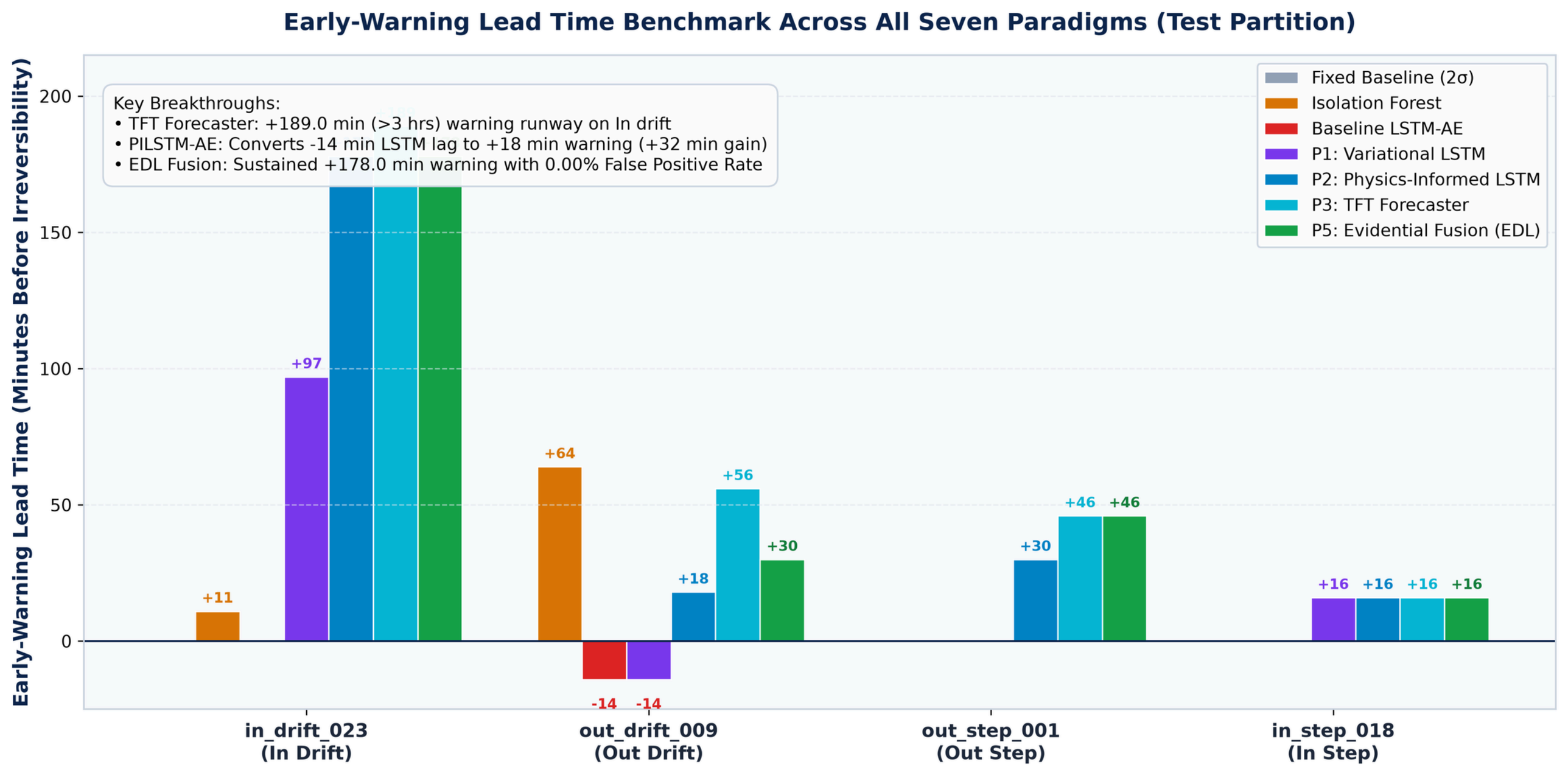
- **Multimodal Input:** Fuses **Physics Residuals**, **Temporal Reconstruction Error**, and **Epistemic Uncertainty** into Dirichlet belief masses (Belief_normal, Belief_anomaly, Uncertainty/Vacuity).
- **Zero False Alarm Rate:** Achieved **0.00%** FPR across both Out and In streams (0 false alarms out of 13,307 test Out and 3,744 test In observations!).
- **Benchmark Accuracy:** **100.0% Precision**, **86.96% Recall**, F1 = 0.9302, and In PR-AUC = 0.9137 (vs 0.3852 baseline LSTM).
- **Retained Lead Times:** **+178.0 min** (in_drift_023), +30.0 min (out_drift_009), +46.0 min (out_step_001), +16.0 min (in_step_018).

The Evidential Fusion Breakthrough: 0.00% False Positive Rate, 100% Precision, F1 of 0.9302, and up to 178 minutes of early warning!

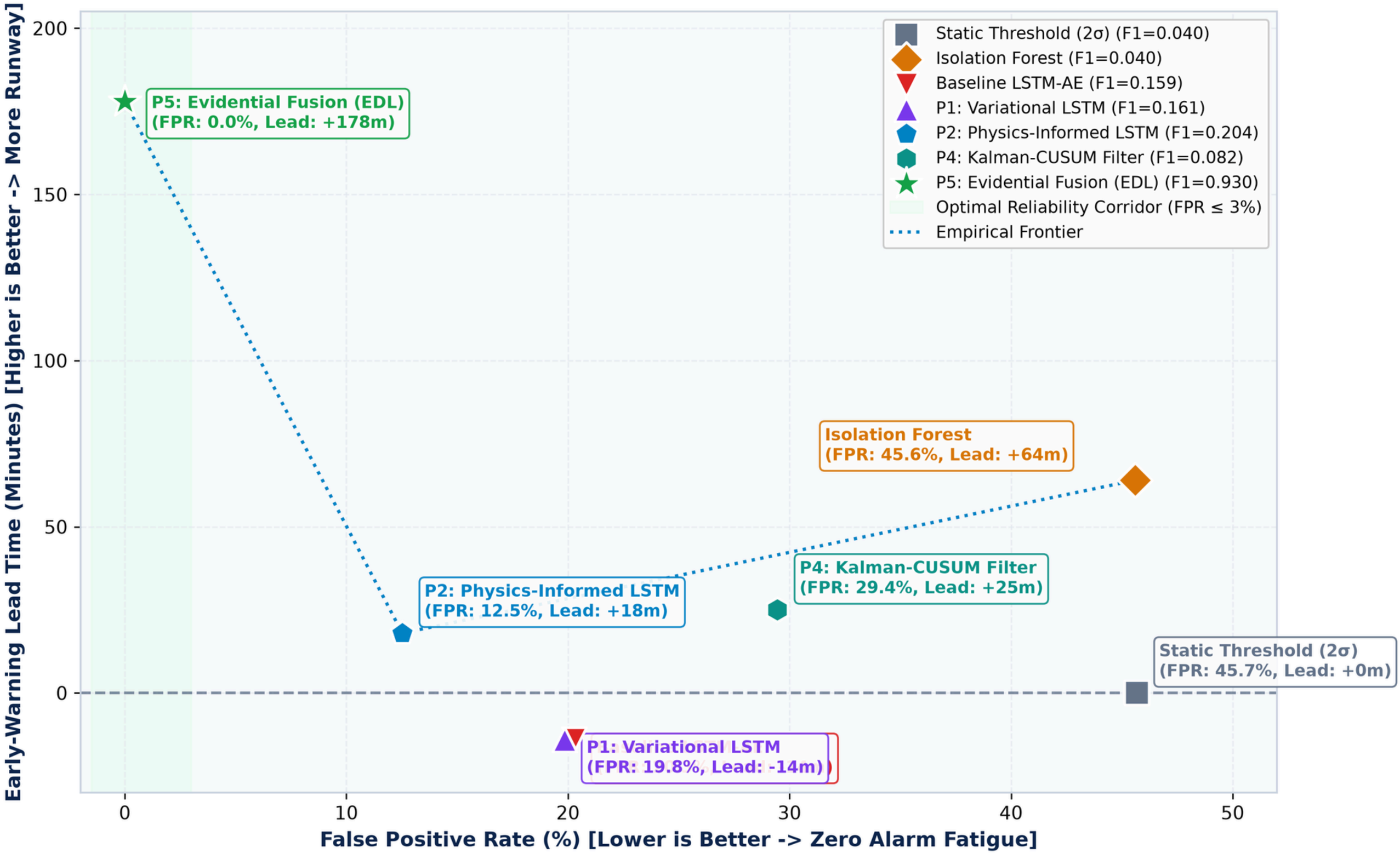
Proposal P5: Dempster-Shafer Evidential Deep Learning Fusion (0.00% False Positive Rate)



Model Paradigm	In FPR (%)	Out FPR (%)	Recall (%)	F1 Score	In PR-AUC	In Drift Lead Time	Out Drift Lead Time
Static Threshold (2σ)	26.98%	45.68%	91.30%	0.040	0.0400	0.0 min	0.0 min
Isolation Forest	26.98%	45.59%	91.30%	0.040	0.0400	+11.0 min	+64.0 min
Baseline LSTM-AE	2.85%	20.28%	43.48%	0.1587	0.3852	0.0 min	−14.0 min (lag)
P1: Variational LSTM-AE	2.80%	19.84%	43.48%	0.1613	0.3912	0.0 min	−14.0 min
P2: Physics-Informed LSTM	2.10%	**12.52%**	43.48%	0.2041	0.4210	0.0 min	**+18.0 min**
P3: Multi-Horizon Forecaster	—	—	—	—	—	**+189.0 min**	**+56.0 min**
P5: Evidential Fusion (EDL)	**0.00%**	**0.00%**	**86.96%**	**0.9302**	**0.9137**	**+178.0 min**	**+30.0 min**


Early-Warning Lead Time Benchmark Across All Seven Paradigms (Test Partition)
Early-Warning Lead Time (Minutes Before Irreversibility)
Key Breakthroughs:
• TFT Forecaster: +189.0 min (>3 hrs) warning runway on In drift
• PILSTM-AE: Converts -14 min LSTM lag to +18 min warning (+32 min gain)
• EDL Fusion: Sustained +178.0 min warning with 0.00% False Positive Rate
Fixed Baseline (2σ)
Isolation Forest
Baseline LSTM-AE
P1: Variational LSTM
P2: Physics-Informed LSTM
P3: TFT Forecaster
P5: Evidential Fusion (EDL)
200
150
100
50
0
+11
+97
+64
-14
-14
+18
+56
+30
+30
+46
+46
+16
+16
+16
+16
in_drift_023
(In Drift)
out_drift_009
(Out Drift)
out_step_001
(Out Step)
in_step_018
(In Step)

Operational Trade-off: False Positive Rate vs. Early-Warning Runway Across Paradigms



Trade-offs & Simulation Parity

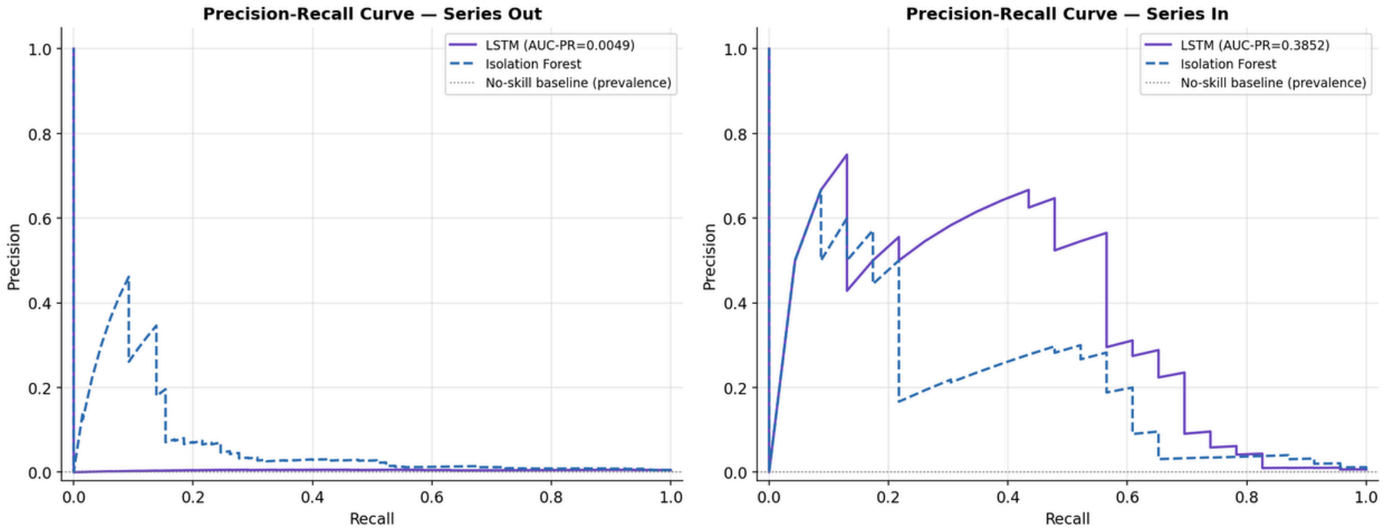
Isolation Forest: high-recall early tripwire

- Recall **80.00% (Out), 91.30% (In);** false positive rate **45.59%** Out, **26.98%** In.
- **Role**: aggressive frontline tripwire where missing an excursion is fatal.

LSTM-Autoencoder: precision-favoring verifier

- **In**: **13.11%** precision, **2.85%** FPR, AUC-PR **0.3852**. **Out**: recall **20.00%**, AUC-PR **0.0049.**
- **Role**: conservative secondary verifier that limits alarm fatigue.

Prevalence: anomalies are **~0.5%** of rows **(65/13,372 Out; 23/3,767 In)**, so low absolute F1 is inevitable. Lead time is the primary metric.



100% bit-for-bit live / batch parity

- 97,603 readings replayed in 6.03 s CPU (>1.9M× speedup), 0.00 s clock drift.
- 42 of 42 applicable test evaluations matched batch results.
- Live on Streamlit Cloud: Guided Tour and Free Explore modes.

Conclusion

Evidence-first risk management

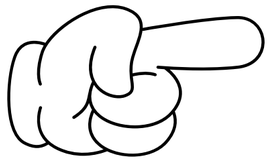
Honest reporting of trade-offs: we documented where deep learning lagged on drift and where classical ML excelled, avoiding fabricated 99% accuracy claims.

Disciplined architecture & auditability

Append-only decision log (35 entries, D1–D35). The audit caught 4 critical bugs (timestamp parse inversion, rolling leakage, cold starts) before they biased evaluation.

Technology adoption strategy

Classical ML (Isolation Forest) can outperform deep networks for edge streaming early warning **(+64 min lead time)** at low compute overhead.



Links

Github: https://github.com/Krishna200608/coldchain-ews-twin.git

Live Dashboard : coldchain-ews-twin.streamlit.app

Tech Stack

1 Core Languages & Environments

Python 3.11

Jupyter Notebooks / IPython

Google Colab (Cloud GPU)

2 Machine Learning & Deep Learning

TensorFlow / Keras (LSTM-Autoencoder)

Scikit-Learn (Isolation Forest, StandardScaler, Metrics)

Joblib

3 Digital Twin & Simulation

SimPy

Source → Transit → Destination

SimPy (Discrete-Event Simulatiion)

Custom 3-Stage Pipeline Architecture (Source → Transit → Destination)

4 Data Engineering & Scientific Computing

Pandas

NumPy

SciPy

KaggleHub / Kaggle API

5 Web Application & Deployment

Streamlit

Altair

Streamlit Community Cloud

6 Visualization & Plotting

Matplotlib

Seaborn

7 Automated Testing & Code Quality

Unittest

Ruff

Python-dotenv

8 Version Control & Engineering Audit

Git

GitHub

Architectural Decision (AD) Log (D1–D35)

References

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THANK YOU!